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BATTERY CHARGING SYSTEM, BATTERY POWER SYSTEM, AND METHOD OF USING QUICK CHARGER TO CHARGE BATTERY OF ELECTRICALLY POWERED TOOL

Abstract

A battery charging system can include each of a heating element, a heating fan, and a cooling fan located in an airflow path, and a controller configured to operate the heating mode when a battery is connected to the battery charging system and a sensed temperature of the battery is less than a predetermined low threshold. The controller can be configured to operate the heating mode by disconnecting a charging power from the battery, supplying a fan power to the heating fan to force air to flow along the airflow path, supplying a heat power to the heating element to emit heat into the airflow path, and disconnecting a cooling power from the cooling fan. In the heating mode, air flowing along the airflow path flows through the heating fan and enters the heating element, and subsequently exits the heating element and passes along the cooling fan.

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Background/Summary

BACKGROUND

[0001] The disclosed subject matter relates to a battery charging system for charging a battery of an electrically powered tool. More particularly, the disclosed subject matter relates to methods and apparatus that heat the battery prior to charging the battery when the battery is cold.

[0002] Power equipment such as, but not limited to, a lawnmower, a snowblower, a lawn edger, a tiller, an aerator, a string trimmer, and a hedge trimmer, can include an electric motor that powers the tool and a rechargeable battery that supplies electrical power to the electric motor. The temperature at which the rechargeable battery is charged can adversely impact the lifespan of the battery.

[0003] A rechargeable battery for power equipment can include a plurality of cells arrayed or stacked inside a housing. The temperature of each of the cells can increase during the charging process. Conventional battery chargers can include a cooling fan that blows air through the housing and along the cells during the charging process so that the temperature of the cells can be maintained below a predetermined maximum temperature.

SUMMARY

[0004] Some embodiments are directed to a battery charging system that can include a heating element located in an airflow path, a heating fan located in the airflow path, a cooling fan located in the airflow path, and a controller configured to operate a heating mode. The heating fan can be located in the airflow path upstream of the heating element when the heating fan is operated during a heating mode. The cooling fan can be located in the airflow path downstream of the heating element when the heating fan is operated during the heating mode. The controller can be configured to operate the heating mode when a battery is connected to the battery charging system and a sensed temperature of the battery is less than a predetermined low threshold. The controller can be configured to operate the heating mode by disconnecting or maintaining disconnection of a charging power from the battery, supplying or maintaining a fan power to the heating fan to force air to flow along the airflow path, supplying or maintaining a heat power to the heating element to emit heat into the airflow path, and disconnecting or maintaining disconnection of a cooling power from the cooling fan. In the heating mode, air flowing along the airflow path flows through the heating fan and enters the heating element, and the air that enters the heating element exits the heating element and passes along the cooling fan.

[0005] Some embodiments are directed to a battery charging system for an apparatus that includes an electric motor and a tool driven by the electric motor. The battery charging system can include a battery and a quick charger. The battery can be configured to be removably mounted on the apparatus and electrically connected to the apparatus to supply electrical power to the motor when the battery is mounted on the apparatus. The quick charger can be connected to a power supply and include a battery dock, a heater, a cooling fan and a controller. The battery dock can be configured to removably receive the battery and electrically connect the battery to the quick charger when the battery is received in the battery dock. The heater can be located on an airflow path that passes along the battery dock and configured to heat the battery when the battery is received in the battery dock. The cooling fan can be located on an airflow path between the battery dock and the heater. The controller can be configured to operate a heating mode when the battery is received in the battery dock and a sensed temperature is less than a predetermined low threshold. The controller can be configured to operate the heating mode by disconnecting or maintaining disconnection of a charging power from the battery, connecting or maintaining connection of a heat power to the heater, and disconnecting or maintaining disconnection of a cooling power from the cooling fan.

[0006] Some embodiments are directed to a method of using a quick charger to charge a battery

mounted in the quick charger. The quick charger can have a heater, a cooling fan, and an opening adjacent to the battery when the battery is mounted in the quick charger. The method can include: locating the heater, the cooling fan and the opening in an airflow path that extends from the heater and through the opening such that the cooling fan is located between the heater and the opening; and when a sensed temperature of the battery is less than a predetermined low threshold, disconnecting or maintaining disconnection of the battery from a power supply; connecting or maintaining connection of the heater to one of the power supply and the battery, and disconnecting or maintaining disconnection of the cooling fan from the power supply.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The disclosed subject matter of the present application will now be described in more detail with reference to exemplary embodiments of the apparatus and method, given by way of example, and with reference to the accompanying drawings, in which:

[0008] FIG. 1 is a perspective view of a battery connected to a battery dock of a battery charging system made in accordance with principles of the disclosed subject matter.

[0009] FIG. 2 is a perspective view of the battery charging system of FIG. 1, with the battery removed from the battery dock of the battery charging system.

[0010] FIG. 3 is a rear perspective view of a snowblower using a battery that can be charged with the battery charging system of FIG. 1.

[0011] FIG. 4 is a schematic illustration of the battery charging system of FIG. 1 with the battery removed from the battery dock.

[0012] FIG. 5 is a schematic illustration of the battery and the battery charging system of FIG. 1, with the battery charging system operating in a heating mode.

[0013] FIG. 6 is a schematic illustration of the battery and the battery charging system of FIG. 1, with the battery charging system operating in a cooling mode.

[0014] FIG. 7 is a flowchart depicting an operation of the battery charging system of FIG. 1.

[0015] FIG. 8 is a perspective cross-sectional view of the battery shown in the battery charging system of FIG. 1.

[0016] FIG. 9 is a schematic illustration of an alternate embodiment of a battery charger system made in accordance with principles of the disclosed subject matter.

[0017] FIG. 10 is an end view of the battery in the battery charging system of FIG. 1.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0018] A few inventive aspects of the disclosed embodiments are explained in detail below with reference to the various figures. Exemplary embodiments are described to illustrate the disclosed subject matter, not to limit its scope, which is defined by the claims. Those of ordinary skill in the art will recognize a number of equivalent variations of the various features provided in the description that follows.

[0019] Some battery charging systems can take many hours or most of a day to charge the battery. Quick battery charging systems can charge the battery to 80% capacity in one hour or less, and charge the battery to 100% capacity in less than five hours.

[0020] Charging a battery with a quick charger at a temperature that is lower than a predetermined minimum temperature can adversely affect the lifespan and/or performance of the battery. Thus, battery's manufacturer sometimes recommends a temperature range for quick charging a battery. The temperature range can be greater than or equal to a predetermined minimum temperature and less than or equal to a predetermined maximum temperature.

[0021] Typically, an owner of power equipment stores the power equipment in an indoor environment that is not heated. Some power equipment such as, but not limited to, snowblowers

can be operated and/or stored in an environment with an ambient temperature that is below the predetermined minimum temperature. Thus, it is possible for an owner or potential customer to perceive power equipment that is operated by an electric battery to be incompatible with their needs if it becomes difficult to charge the battery at low temperatures.

[0022] Further, the owner or potential customer may adversely perceive the benefit of a quick charger when the battery is stored or used in an environment with an ambient temperature that is below the predetermined minimum temperature. Although the battery could be stored or brought into a heated environment, the owner of the power equipment can perceive the storage of the battery in the heated environment as inconvenient. Alternatively, the owner or user could bring the battery that has been stored in a relatively cold environment into a relative warm environment to warm the battery. The time to raise the battery temperature above the minimum threshold can be longer than the charging time when using a quick charger. Thus, the owner or user of the quick charger can negatively perceive the benefit of the quick charger.

[0023] Accordingly, there is a need for a battery charging system that can relatively quickly charge a battery that is initially at a temperature that is below the predetermined minimum temperature.

[0024] FIG. 1 illustrates an embodiment of a battery charging system **10** made in accordance with principles of the disclosed subject matter. The battery charging system **10** can use a process that allows the battery charging system **10** to perform a quick charge cycle. In certain cases, the battery charging system **10** can use a process that performs a charge cycle that is longer in time than a quick charge cycle. The battery charging system **10** can charge a battery **12** for use in electrically powered equipment **14** (see FIG. 3) such as but not limited to a lawnmower, a snowblower, a lawn edger, a tiller, an aerator, a string trimmer, and a hedge trimmer. The electrically powered equipment can be referred to as power equipment, a power tool, a powered tool, an electric tool, a power machine, a powered machine, or an electric machine. The battery charging system **10** and the battery **12** can be collectively referred to as a battery power system.

[0025] FIG. 2 shows the battery charging system **10** with the battery **12** removed. FIG. 3 shows an exemplary electrically powered equipment **14** configured as a snowblower that can use the battery **12** that has been charged by the battery charging system **10**. The battery **12** can be removed from the snowblower **14** and charged in the battery charging system **10** and then returned to the snowblower **14** or used in a different type of electrically powered equipment that is compatible with the battery **12**.

[0026] FIG. 4 schematically illustrates the battery charging system **10** of FIG. 1. The battery charging system **10** can include a heater **16** that heats the battery **12** when the battery **12** is less than a predetermined low temperature threshold. The battery charging system **10** can be configured to automatically start charging the battery **12** when the temperature of the battery **12** is greater than or equal to the low temperature threshold. The predetermined low temperature threshold can be greater than or equal to the predetermined minimum temperature that is set by the manufacturer of the battery **12**. The heater **16** can be configured to transfer heat to the battery **12** at a rate that is sufficient for the owner or user of the battery charging system **10** to positively perceive the time that has elapsed to heat and charge the battery **12**.

[0027] Returning to FIGS. 1 and 2, the battery charging system **10** can include a housing **18** and a battery dock **20** that is connected to the housing **18** in any appropriate manner such as, but not limited to, threaded fasteners, elastic clips, clamps, staking, welding, adhesive. A seam **38** can delineate the housing **18** from the battery dock **20**. The seam **38** can extend around the entire perimeter of the housing **18** and the battery dock **20**.

[0028] Referring to FIG. 2, the battery dock **20** can include a connector **22**, an end wall **26**, a support wall **28**, and a pair of side walls **30**, **32**. The connector **22** is obstructed from view in FIG. 1 by the first side wall **30** and the battery **12** and obstructed from view in FIG. 2 by the first side wall **30**. The connector **22** is schematically illustrated in phantom in FIG. 2 and schematically illustrated in FIG. 4. Referring to FIGS. 5, 6, and 10, the battery **12** can include a mating connector **24** that

can be connected to the connector **22** to electrically connect the battery **12** to the battery charging system **10**. The mating connector **24** of the battery **12** is obstructed from view in FIG. **1** by the first side wall **30** and the battery **12**. The mating connector **24** is schematically illustrated in FIGS. **5**, **6**, and **10**.

[0029] Returning to FIG. **2**, the connector **22** can protrude from the support wall **28**. Each of the connectors **22**, **24** can include one or more terminals for transmitting charging power and one or more terminals for transmitting electrical control signals between the battery **12** and battery charging system **10**. Each of the power terminal(s) of the connector **22** can be connected to a respective one of the power terminal(s) of the mating connector **24** and each of the signal terminal(s) of the connector **22** can be connected to a respective one of the signal terminal(s) of the mating connector **24**.

[0030] The walls **26**, **30**, **32** can extend away from the support wall **28**. The walls **26**, **28**, **30**, **32** can extend along and form a boundary of an external space **40** of the battery dock **20**. That is, the external space **40** can extend along the outer surfaces of the battery dock **20** that are located on the walls **26**, **28**, **30**, **32**. The battery **12** can be located in the external space **40** when the battery **12** is mounted on or in the battery dock **20** with the connectors **22**, **24** connected to each other.

[0031] The first end wall **26** can include an end surface **36** that faces the battery **12** and is adjacent to the battery **12** when the battery **12** is mounted in the battery dock **20**. Referring to FIGS. **1** and **10**, the battery **12** can include a first end surface **42** that faces the end surface **36** of the battery dock **20**. The surfaces **36**, **42** can be spaced away from each other.

[0032] Referring to FIG. **4**, the first end wall **26** can be a hollow wall that includes an internal space **52**. The battery dock **20** can include a dock opening **54** extending through the end surface **36**. The dock opening can be in fluid communication with each of the external space **40** and the internal space **52**. Referring to FIG. **2**, the dock opening **54** can include a plurality of holes **58** extending through the end surface **36**.

[0033] Referring to FIGS. **1**, **2** and **4** collectively, the housing **18** can include a first end wall **44**, a second end wall **34**, a pair of side walls **46**, **48** and a bottom wall **50**. Referring to FIG. **4**, the housing **18** can include an interior space **60** that is bound by the walls **34**, **44**, **46**, **48**, **50**. The housing **18** can include an opening that is bound by edges of the walls **34**, **44**, **46**, **48** that are adjacent to the seam **38**. Components of the battery charging system **10** can be inserted into the interior space via the opening during assembly of the battery charging system **10**. The support wall **28** can extend along the internal space **52** and close the opening. The interior space **60** can be referred to as a compartment.

[0034] Referring to FIG. **4**, the battery charging system **10** can include a cooling fan **62**. The cooling fan **62** can be located in the internal space **52** of the end wall **26**. The cooling fan **62** can be any appropriate fan such as, but not limited to, an axial-flow fan or a centrifugal fan. The internal space **52** can be referred to as a compartment.

[0035] The battery dock **20** can include an intermediate opening **66** that extends through the support wall **28**. The intermediate opening **66** can be in fluid communication with both of the internal space **52** of the battery dock **20** and the interior space **60** of the housing **18**.

[0036] Returning to FIGS. **1**, **2** and **4**, the housing **18** can include housing opening **68** that is in fluid communication with the interior space **60** of the housing **18** and an ambient environment that is outside of the housing **18** and the battery dock **20**. The housing opening **68** can extend through the first side wall **30** as shown in FIGS. **1** and **2**. In alternate embodiments, the housing opening **68** can extend through the bottom wall **50** as schematically illustrated in FIGS. **4-6**. Referring to FIGS. **1** and **2**, the housing **18** can include a grille **70** that extends across the housing opening **68**.

[0037] Referring to FIG. **4**, each of the openings **54**, **66**, **68**, the cooling fan **62** and the heater **16** can be located on an airflow path **A1** that is schematically indicated by a double headed arrow. The heater **16** and the airflow path **A1** can be configured to draw air from the ambient environment outside of the housing **18** into the housing **18** via the housing opening **68** and exhaust the heated air

through the dock opening **54** and into the external space **40** of the battery dock **20**. The cooling fan **62** and the airflow path **A1** can be configured to draw air from the external space **40** through the dock opening **54** and into the housing **18** and exhaust the air via the housing opening **68** to the ambient environment.

[0038] Referring to FIG. **8**, the battery **12** can include a battery housing **70**, a plurality of battery cells **72** and a plurality of sensors **74**. The inside of the housing **70** can be referred to as a battery compartment **84** and the battery cells **72** can be contained in the battery compartment **84**. Each of the sensors **74** can be mounted on a respective one of the battery cells **72**.

[0039] The battery cells **72** can be arranged in any appropriate manner such as, but not limited to, a plurality of staggered rows or staggered columns. The battery cells **72** can include a middle battery cell **72m** and an end battery cell **72e**. The middle battery cell **72m** can be encircled by a subset of the plurality of battery cells **72**. FIG. **8** shows a subset of six battery cells **72** encircling the middle battery cell **72m**. The end battery cell **72e** can be the last battery cell **72** of a given row or given column of the battery cells **72**.

[0040] Each of the sensors **74** can be any appropriate temperature sensor such as, but not limited to a thermistor. Electrical wires **94** can connect the sensors **72** to the electronic component **76**. The sensors **74** can be configured to output data (or a signal) that is indicative of the temperature of the associated one of the battery cells **72e**, **72m**. The electronic component **76** can be configured to store the temperature data until requested by the controller **64** and transmit the temperature data to the controller **64** when requested by the controller **64**.

[0041] Referring to FIGS. **1**, **8** and **10** collectively, the battery housing **70** can include a first ventilation opening **78**, a second end wall **80**, a second ventilation opening **82** and a battery compartment **84**. The first ventilation opening **78** can extend through the first end surface **42**. The first ventilation opening **78** can be located in the external space **40** of the battery dock **20** and oppose the dock opening **54** when the connectors **22**, **24** are connected to each other. The second end **80** wall can be spaced away from the first end surface **42** and the second ventilation opening **82** can extend through the second end wall **80**. Both of the second end wall **80** and the second ventilation opening **82** can be outside of both of the housing **18** and the battery dock **20**. Both of the ventilation openings **78**, **82** can be in fluid communication with the battery compartment **84**. The first ventilation opening **78** can include a plurality of holes **86** extending through the first end surface **42** and the second ventilation opening **82** can include a plurality of holes **88** extending through the second end wall **80**.

[0042] The battery charging system **10** can heat the battery **12**, charge the battery **12**, and cool the battery **12**, when the battery **12** is mounted on or in the battery dock **20** and the mating connector **24** is connected to the connector **22**. FIG. **5** shows air flowing along the airflow path **A1** when the battery charging system **10** is operating in a heating mode that heats the battery **12** prior to charging the battery **12**. FIG. **6** shows air flowing along the airflow path **A1** when the battery charging system **10** is operating in a cooling mode that cools the battery **12** while charging the battery **12**. That is, the battery **12** can remain mounted on or in the battery dock **20** when the battery charging system **10** heats the battery **12**, charges the battery **12**, and cools the battery **12** during charging. The cooling fan **62** is omitted from FIG. **5** and the heater **16** is omitted from FIG. **6** for clarity and simplicity of the drawings.

[0043] Returning to FIG. **4**, the heater **16** can include a heating fan **90** and a heating element **92**. The heating fan **90** can be any appropriate fan such as, but not limited to, an axial-flow fan or a centrifugal fan. The heating element **92** can be any appropriate heating device such as, but not limited to, an electric resistance heating element, a ceramic heating element, or an infrared heat lamp, or any combination of these exemplary heating devices.

[0044] The heating fan **90** and the heating element **92** can be located in the airflow path **A1** between the intermediate opening **66** and the housing opening **68**. The heating fan **90** can be positioned in the airflow path **A1** at a location that is between the heating element **92** and the

housing opening **68**. The heating element **92** can be positioned in the airflow path **A1** at a location that is between the heating fan **90** and the intermediate opening **66**. That is, the heating fan **90** can be spaced away from the housing opening **68** by a distance that is less than the distance by which the heating element **92** is spaced from the housing opening **68**, and the heating element **92** can be spaced away from the intermediate opening **66** by a distance that is less than the distance by which the heating fan **90** is spaced from the intermediate opening **66**, as shown in FIG. 4.

[0045] The battery charging system **10** can operate in a heating mode when the temperature of the battery **12** is less than or equal to a predetermined low temperature $T_{sub.Low}$ and in a cooling mode when the temperature of the battery **12** is greater than or equal to a predetermined high temperature $T_{sub.High}$. The predetermined low temperature $T_{sub.Low}$ can be greater than or equal the predetermined minimum temperature set by the manufacture of the battery **12**. The predetermined high temperature $T_{sub.High}$ can be less than or equal the predetermined maximum temperature set by the manufacture of the battery **12**.

[0046] When the battery charging system **10** and the heater **16** are in a heating mode, air from the ambient environment outside of the housing **18** can be drawn into the housing **18** by the heater **16** and the air heated by the heater **16** can be exhausted into the battery **12** as indicated by arrows **H1-H4** in FIG. 5. The heating fan **90** can draw the ambient air that is outside the housing **18** through the housing opening **68** and into the interior space **60** of into the housing **18** as indicated by the first arrow **H1**. The heating fan **90** can force the air that is drawn in through the housing opening **68** and into the housing's interior space **60** to flow across and/or through the heating element **92**. The heating fan **90** can force the air flowing across the heating element **92** to exit the interior space **60** through the intermediate opening **66** and enter the internal space **52** of the end wall **26** of the battery dock **20** as indicated by the second arrow **H2**. The heating fan **90** can force the air that enters the internal space **52** via the intermediate opening **66** to exit the internal space **52** through the dock opening **54** and enter the battery **12** via the first ventilation opening **78** as indicated by the third arrow **H3**. The heating fan **90** can force the air that enters the battery **12** to flow through the battery compartment **84** and across or along each of the battery cells **72**. The heating fan **90** can force the air flowing across or along the battery cells **72** to exit the battery compartment **84** through the second ventilation opening **82** and flow into the ambient environment as indicated by the fourth arrow **H4**. Thus, the heater **16** can heat the battery cells **72** when the battery cells **72** are at a temperature that is less than the predetermined low temperature $T_{sub.Low}$. The battery charging system **10** can be configured to automatically charge the battery **12** after the battery charging system **10** heats the battery to a temperature that is greater than or equal to the predetermined low temperature $T_{sub.Low}$.

[0047] During charging, the temperature of the battery **12** can increase. Quick charging (also referred to as fast charging) the battery **12** when the battery **12** is at a temperature that is greater than a predetermined maximum temperature can adversely impact the battery **12**. Thus, the battery charging system **10** and the cooling fan **62** can operate in the cooling mode to maintain the battery temperature at or below the predetermined high temperature $T_{sub.High}$.

[0048] When the battery charging system **10** and the cooling fan **62** are in a cooling mode, air from the ambient environment outside of both of the battery **12** and the battery charging system **10** can be drawn through the battery **12** by the cooling fan **62** as indicated by arrows **C5-C8** in FIG. 6. The cooling fan **62** can draw the ambient air that is outside the battery **12** through the second ventilation opening **82** and into the battery compartment **84** as indicated by the fifth arrow **C5**. The cooling fan **62** can force the air that is in the battery compartment **84** to flow across or along each of the battery cells **72**. The cooling fan **62** can force the air flowing across the battery cells **72** to exit the battery compartment **84** through the first ventilation opening **78** and enter the internal space **52** of the end wall **26** of the battery dock **20** via the dock opening **54** as indicated by the sixth arrow **C6**. The cooling fan **62** can force air in the internal space **52** to exit the internal space **52** through the intermediate opening **66** and enter the interior space **60** of the housing **18** as indicated by the

seventh arrow **C7**. The cooling fan **62** can force the air to exit the interior space **60** through the housing opening **68** and into the ambient environment. Thus, the cooling fan **62** can cool the battery cells **72** when the battery cells **72** are at a temperature that is greater than the predetermined high temperature $T_{sub.High}$.

[0049] Returning to FIG. **4**, the battery charging system **10** can include a controller **64** in electrical communication with each of the heater **16**, the connector **22** and the cooling fan **62** as shown schematically by the dotted lines. Each of the dotted lines can schematically represent at least one power supply line and/or at least one electrical communication line. The controller **64** can be referred to as a processor based controller, a microcomputer, an electronic control unit, or an ECU. The controller **64** can be configured with hardware alone, or with a combination of hardware and software that permit the controller **64** to perform the tasks described herein. The controller **64** can include a central processing unit ("CPU"), a RAM, a ROM and/or an interface that connects the controller **64** to any or all of the connector **22**, the heater **16** and the cooling fan **62**. The controller **64** can be connected to an external storage device or include an internal storage device in which software can be stored and accessed by the CPU.

[0050] Referring to FIGS. **5** and **6**, the battery **12** can include an electronic component **76** that is in electrical communication with each of the mating connector **24** and the sensors **74**. The electronic component **76** can be a processor-based controller, and/or a memory storage device, and/or a communication interface. The electronic component **76** can be configured to receive data from the sensors **74** and transmit the data to the controller **64** via the connectors **22**, **24**.

[0051] When the electronic component **76** is configured as a controller, the electronic component **76** can be configured with hardware alone, or with a combination of hardware and software that permit the electronic component **76** to perform the tasks described herein. The electronic component **76** can include a central processing unit ("CPU"), a RAM, a ROM and/or an interface that connects the electronic component **76** to any or all of the mating connector **24** and the sensors **74**.

[0052] The battery charging system **10** can be connected to an external power supply **200** in any appropriate manner such as, but not limited to, a plug and wire assembly that can be selectively plugged into and removable from each of the battery charging system **10** and a wall outlet, a plug and wire assembly that is non-removably connected to the battery charging system **10** and selectively plugged into and removable from the wall outlet, or permanent electrical connection. The battery charging system **10** can include an inverter when the external power supply is an alternating current power supply.

[0053] The battery charging system **10** can include a respective power supply circuit for each of the heater **16**, the cooling fan **62** and charging the battery **12**. Each power supply circuit can modulate an output voltage. The power supply circuits can be part of the controller **64** or separate and distinct from the controller **64**. The controller **64** can be connected to the external power supply **200**. The controller **64** can include or be electrically connected to a respective power supply circuit for each of the heater **16**, the battery operation and the cooling fan **62**. Each electrical circuit can include one or more components such as, but not limited to, a voltage converter or a power MOSFET that can convert power received from the external power supply **200** to a respective driving power for each heater **16** and the cooling fan **62** and a charging power for the battery **12**. The power MOSFET can be manipulated by the controller **64** between an OFF state and an ON state. When any of the powered switches is in the ON state, the respective switch can connect electrical power to respective one(s) of the heater **16**, the cooling fan **62** and the battery **12**. Each of the power supply circuits can be configured to derive a respective one of a driving power and a charging power from the power received from the external power supply **200**.

[0054] The controller **64** can be configured to set a respective driving power value for each of the fans **62**, **90** and the heating element **92** so that the respective power circuit outputs electrical power derived from the external power supply **200** at the corresponding driving power value. In some

embodiments, each of the driving power values can be a respective predetermined fixed value. In other embodiments, the controller **64** can be configured to vary, as a function of the temperature $T_{sub.batt}$ of the battery **12**, any one or both of the driving power values.

[0055] The controller **64** can be configured to set a respective charging power value for battery **12** so that the respective power circuit outputs electrical power derived from the external power supply **200** at the corresponding charging power value. In some embodiments, the controller **64** can be configured to vary the charging power as a function of the state of charge (“SoC”) of the battery **12**. In alternate embodiments, the controller **64** can be configured to set a relatively high voltage power value and subsequently charge the battery **12** with a relatively low current power. The controller **64** can be configured to charge the battery **12** based on the relative high voltage power when the SoC is less than a changeover threshold. The changeover threshold can be a predetermined SoC value that is less than 100% and at which high voltage charging can adversely impact the performance of the battery **12**. The controller **64** can be configured to charge the battery **12** based on the relatively low current power when the SoC of the battery **12** is greater than the changeover threshold and less than 100%.

[0056] The controller **64** be configured to automatically start and stop the heating mode, the charging mode and the cooling mode of the battery charging system **10**. The fans **62**, **90** and the heating element **92** can be in an ON state when the controller **64** supplies or causes the supply of the driving power and in an OFF state when the controller **64** disconnects or causes disconnection of the driving power. The cooling mode can be in an ON state when the cooling fan **62** is in the ON state and each of the heating fan **90** and the heating element **92** are in the OFF state. The heating mode can be in an ON state when the cooling fan **62** is in the OFF state and each of the heating fan **90** and the heating element **92** are in the ON state. Each of the fans **62**, **90** and the heating element **92** can be described as being connected to the external power supply **200** when in the ON state and disconnected from the external power supply **200** when in the OFF state.

[0057] The charging mode can be in an ON state when the controller **64** supplies or causes the supply of the charging power to the battery **12** and in an OFF state when the controller disconnects or causes disconnection of the charging power. The cooling mode can be in the ON state or the OFF state when the charging mode is in the ON state. The heating mode can be in the OFF state when the charging mode is in the ON state, and the heating mode can be in the ON state when the charging mode is in the OFF state. The battery **12** can be described as being connected to the external power supply **200** when the charging mode is in the ON state and disconnected from the external power supply **200** when the charging mode is in the OFF state.

[0058] FIG. 7 illustrates a flowchart that represents a battery heating and charging algorithm that can be implemented by the controller **64** in order to start and stop each of the modes of the battery charging system **10**.

[0059] The controller **64** can be configured to enter the battery heating and charging algorithm at step **S100**. The controller **64** can be configured to detect the presence or absence of the connection between the connectors **22**, **24**, and initiate step **S100** when the controller **64** determines that the connection between the connectors **22**, **24** is present. From step **S100**, the controller **64** can proceed to step **S102**.

[0060] At step **S102**, the controller **64** can be configured to determine the SoC of the battery **12** and obtain the battery temperature $T_{sub.batt}$. The controller **64** can be configured to determine the SoC by any appropriate method that is currently known such as, but not limited to, using the specific gravity of the electrolyte, using the battery voltage and a predetermined discharge curve, or using current integration, or using any appropriate method developed in the future. The controller **64** can be configured to obtain the battery temperature $T_{sub.batt}$ based on the temperature data (or signal) output by the sensors **74**.

[0061] At step **S102**, the controller **64** can be configured to transmit a request to the electronic component **76** to transmit the temperature data (or signal) received from the sensors **74**. In alternate

embodiments, the electronic component **76** can be configured to transmit the temperature data (or signal) to the controller **64** at predetermined intervals. After the controller **64** obtains the temperature T.sub.batt and determines the SoC of the battery **12**, the controller **64** can proceed to step **S104**.

[0062] At step **S104**, the controller **64** can be configured to determine whether the SoC corresponds to a 100% charged state of the battery **12**. The battery **12** can be considered to be fully charged when the SoC is equal to 100% and in a chargeable state when the SoC is less than 100%. If the controller **64** determines that SoC is less than 100%, the controller **64** can proceed to step **S106**. If the controller **64** determines that the SoC is not less than 100%, the controller **64** can proceed to step **S118**.

[0063] If the controller **64** proceeds from step **S104** to step **S106**, the controller **64** can be configured to determine whether or not the battery temperature T.sub.Batt is acceptable for charging the battery **12**. At step **S106**, the controller **64** can be configured to compare the battery temperature T.sub.Batt to the predetermined low temperature T.sub.Low. If the battery temperature T.sub.Batt is less than the predetermined low temperature T.sub.Low, then the battery **12** can be near or below the predetermined minimum temperature and the battery charging system **10** and the heater **16** can be used to heat the battery **12** before the battery charging system **10** charges the battery **12**. If the battery temperature T.sub.Batt is less than the predetermined low temperature T.sub.Low, then the controller **64** can proceed to step **S108**. If the battery temperature T.sub.Batt is greater than or equal to the predetermined low temperature T.sub.Low, then the controller **64** can proceed to step **S112**.

[0064] If the controller **64** proceeds from step **S106** to step **S108**, then the battery **12** is too cold for charging. At step **S108**, the controller **64** can be configured to initiate the heating mode of the battery charging system **10**, or maintain the heating mode in the ON state if the heating mode is presently in the ON state. The controller **64** can be configured to disconnect or maintain disconnection of the cooling fan **16** from its respective driving power (i.e., cooling fan **62** is in the OFF state), and disconnect or maintain disconnection of the battery **12** from its charging power (i.e., charging mode is in OFF state). The controller **64** can be configured to set and output (or cause the output of) the driving power for the heating fan **90** and set and output (or cause the output of) the driving power for the heating element **92**. The controller **64** can be configured to proceed to step **S110** after completing step **S116**.

[0065] At step **S110**, the controller **64** can be configured to exit the battery heating and charging algorithm. The controller **64** can be configured to execute the battery heating and charging algorithm at predetermined intervals of operation of the battery charging systems **10**.

[0066] If the controller **64** proceeds from step **S106** to step **S112**, the battery temperature T.sub.Batt is acceptable for charging the battery **12**. The battery temperature T.sub.Batt can increase during charging. If the battery temperature T.sub.Batt is greater than the predetermined high temperature T.sub.High, then the battery **12** is approaching the predetermined maximum temperature and the battery charging system **10** can be operated in the cooling mode (i.e., the cooling mode is in the ON state). If the battery temperature T.sub.Batt is less than or equal to the predetermined high temperature T.sub.High, then the battery **12** is in an acceptable state for charging without the cooling mode in the ON state.

[0067] At step **S112**, the controller **64** can be configured to determine whether the cooling mode should be placed in the ON state. The controller **64** can be configured to compare the battery temperature T.sub.Batt to the predetermined high temperature T.sub.High. If the controller **64** determines that the battery temperature T.sub.Batt is less than or equal to the predetermined high temperature T.sub.High, then the controller **64** can be configured to proceed to step **S114**. If the controller **64** determines that the battery temperature T.sub.Batt is not less than or equal to the predetermined high temperature T.sub.High, then the controller **64** can be configured to proceed to step **S116**.

[0068] At step S114, the controller **64** can start or maintain the charging mode in the ON state, and each of the heating mode and the cooling mode in the OFF state. The controller **64** can be configured to connect or maintain connection of the battery **12** to the charging power, disconnect or maintain disconnection of the heater **16** from its respective driving power (i.e., heater **16** is in the OFF state), and disconnect or maintain disconnection of the cooling fan **62** to its respective driving power (i.e., cooling fan **62** is in the OFF state). The controller **64** can be configured to set and output (or cause the output) of the charging power for the battery **12**. The controller **64** can be configured to proceed to step S110 after completing step S114 and exit the battery heating and charging algorithm.

[0069] If the controller **64** proceeds from step S112 to step S116, then controller **64** can be configured to start the cooling mode and the charging mode of the battery charging system **10**, or maintain any of the cooling mode and the charging mode in the ON state if any of the charging mode and the cooling mode is presently in the ON state. The controller **64** can be configured to disconnect or maintain disconnection of the heater **16** from its respective driving power (i.e., heater **16** is in the OFF state), connect or maintain connection of the cooling fan **62** to its respective driving power (i.e., cooling fan **62** is in the ON state), and connect or maintain connection of the battery **12** to its charging power (i.e., charging mode is in the ON state). The controller **64** can be configured to set and output (or cause the output) of the driving power for the cooling fan **62** and set and output (or cause the output) of the charging power for the battery **12**. The controller **64** can be configured to proceed to step S110 after completing step S116 and exit the battery heating and charging algorithm.

[0070] If the controller **64** proceeds from step S104 to step S118, then the battery **12** is fully charged (i.e., the SoC is equal to 100%) and the controller **64** can stop operating the battery charging system **10**. At step S118, the controller **64** can be configured to turn off the heater **16**, turn off the charging mode, and turn off the cooling fan **62**. That is, the controller can be configured to place each of heater **16**, the charging mode, and the cooling fan **62** in the OFF state. In the event that any of the heater **16**, the charging mode and the cooling fan are in the OFF state when the controller **64** enters step S118, the controller **64** can be configured to maintain the OFF state of any of the heater **16**, the charging mode and the cooling fan **62** that are presently in the OFF state. That is, controller **64** can be configured disconnect or maintain disconnection of each of the heater **16**, the charging mode and the cooling fan **62** from the external power supply **200**. After completing step S118, the controller **64** can proceed to step S110 and exit the battery heating and charging algorithm.

[0071] Thus, the battery charging system **10** can charge a battery **12** that is initially too cold for charging. The battery charging system **10** can automatically detect the temperature of the battery **12** and determine the ON state or the OFF state for each the heating mode, the charging mode and the cooling mode. The automatic selection of the heating mode, the charging mode and the cooling mode can be perceived as convenient and efficient by the owner or user of the battery charging system **10**.

[0072] In order enhance the positive perception by the owner or user, the battery charging system **10** can provide the owner or user with information regarding the operation of battery charging system **10**. Returning to FIG. 4, the battery charging system **10** can include a display **96** that is in electrical communication with the controller **64**. The display **96** can be any appropriate display such as, but not limited to, a touch screen or a liquid crystal display (“LCD”). The display can be configured to display any one or any combination of text, image(s), or icon(s), in color or monochrome. The text, image(s), icon(s) can convey information indicating any one of or any combination of the SoC, the battery temperature T.sub.Batt, ambient temperature, the heat output by the heater **16**, the remaining time for heating the battery **12**, the remaining time for charging the battery **12** and/or any other information that is pertinent to the charging of the battery **12**, usage of the battery **12**, and usage of the battery charging system **10**.

[0073] The controller **64** can be configured to calculate the remaining time for heating battery **12** based on the temperature data from the sensors **74** and the driving power set for heating fan **90** and the heating element **92**. The controller **64** can be configured to output a signal to the display **96**, or a separate driver for the display **96**, that causes the display **96** to display the remaining time. The controller **64** can be configured to update the displayed information in predetermined intervals. The controller **64** can be configured to cause the display **96** to indicate when the battery **12** is fully charged. Thus, the display **96** can enhance the owner's or user's positive perception of the battery charging system **10**.

[0074] FIG. **9** schematically illustrates an alternate embodiment of a battery charging system **100** that can include the housing **18**, the battery dock **20**, the cooling fan **62**, the heating element **92** and the display **96** described above with respect to FIGS. **1-8**. The heating fan **90** can be omitted from the heater **16**. Thus, instead of the forced convection of the battery charging system **10**, the battery charging system **100** can use natural convection to transfer heat from the heating element **92** to the battery cells **72**.

[0075] The air inside the interior space **60** of the housing **18** that has been heated by the heating element **92** is warmer and less dense than the air that is outside of the housing **18**, in the battery dock **20**, and in the battery compartment **84**. This density difference can create a convective current that follows the airflow path indicated by the arrows H9-H12. When the battery charging system **100** and the heating element **92** operate in heating mode, the denser ambient air that is outside the housing **18** can enter the housing **18** through the housing opening **68** and flow into the interior space **60** of the housing **18** as indicated by the ninth arrow H9. The air heated by heating element **92** can be drawn through the intermediate opening **66** and enter the internal space **52** of the end wall **26** of the battery dock **20** as indicated by the tenth arrow H10. The heated air in the internal space **52** can exit the internal space **52** through the dock opening **54** and enter the battery **12** via the first ventilation opening **78** as indicated by the eleventh arrow H11. The heating air can flow through the battery compartment **84** and across or along each of the battery cells **72**. The air flowing across or along the battery cells **12** can exit the battery compartment **84** through the second ventilation opening **82** and flow into the ambient environment as indicated by the twelfth arrow H12. Thus, the heater **16** can heat the battery cells **72** when the battery cells **72** are at a temperature that is less than the predetermined low temperature T.sub.LOW.

[0076] The controller **64** of the battery charging system **100** can be configured to execute the battery heating and charging algorithm as described above with respect to FIG. **7**, with the exception that the controller **64** does not set or output (or cause the output of) a driving power for the omitted heating fan **90**.

[0077] Without the heating fan **90**, the battery charging system **100** can weigh less, be less expensive, and be as convenient as compared to the battery charging system **10**.

[0078] While certain embodiments of the invention are described above, it should be understood that the invention can be embodied and configured in many different ways without departing from the spirit and scope of the invention.

[0079] Instead of deriving the driving power for the heating element **92** from the external power supply, alternate embodiments of the battery charging system **10** or **100** can be configured to supply the driving power from the battery **12**. The battery temperature T.sub.Batt of the battery **12** can increase when the battery **12** discharges electric power to a load. Due to the relative high resistance of the heating element **92**, the heat generated in the battery **12** can be of a rate that is advantageous for heating the battery **12** when the battery temperature T.sub.Batt is less than the predetermined low temperature T.sub.Low. Further, the convective heating created by the heating element **92** alone (FIG. **9**) or in combination with the heating fan **90** can reduce the elapsed time for the battery temperature T.sub.Batt to exceed the predetermined low temperature T.sub.Low.

[0080] In the alternate embodiments in which the battery **12** supplies power to the heating element **92**, the controller **64** or the electronic component **76** can be configured to selectively connect and

disconnect the battery **12** to/from the heating element **92** based on the battery temperature T.sub.Batt, the SoC of the battery **12**, and/or any other appropriate parameter(s).

[0081] In the embodiments of FIGS. **1-9**, the controller **64** can be configured to execute all of the steps of the battery heating and charging algorithm of FIG. **7**. However, alternate embodiments can split the steps amongst the controller **64** and the electronic component **76** where the electronic component **76** is a second controller. In exemplary embodiments, the electronic component **76** can be configured to perform steps **S100**, **S102**, **S106** and **S112** and transmit appropriate data (or signal(s)) to the controller **64**, and the controller **64** can be configured to perform steps **S100**, **S108**, **S114**, **S116** and **S118** and request the appropriate data (or signal(s)) from the electronic component **76**. However, either the controller **64** or the electronic component **76** as a second controller can execute any one of the steps or any subcombination of the steps of FIG. **7** and the other of the controller **64** and the electronic component **76** as a second controller can execute any of the remainder of the steps of FIG. **7**. Alternatively, a single controller can be used in place of the disclosed controller **64** and electronic component **76**, or multiple controllers and electronic parts can be used.

[0082] In alternate embodiments, the controller **64** and/or the electronic component **76** can execute the steps **S104-S108** and **S112-S118** in any order that is different from that shown in and described with reference to FIG. **7**, so long as the controller **64** and/or the electronic component **76** can properly cause the heating, charging and cooling of the battery **12**.

[0083] In alternate embodiments, the electronic component **76** can be omitted and the controller **64** can be configured to receive the data (or signals) from the sensors **74** via the connectors **22**, **24**.

[0084] Instead of charging the battery **12** via electricity flowing from the connector **22** to the mating connector **24**, the battery **12** and the battery charging system **10** can be configured for wireless charging (and/or wireless communication between components). Instead of the connectors **22**, **24**, the battery **12** and the battery dock can have appropriate wireless charging structures.

[0085] In alternate embodiments, the electronic component **76** can be omitted and all of the tasks described with reference to FIG. **7** can be performed by the controller **64**.

[0086] In alternate embodiments, the controller **64** can be configured to initially set the driving power for the cooling fan **62** at a maximum value and then decrease the driving power as a difference between the battery temperature T.sub.Batt and the predetermined high temperature T.sub.High decreases.

[0087] In alternate embodiments, the controller **64** can be configured to initially set the driving power for the heating fan **62** and the driving power for the heating element **92** at respective maximum values and then decrease the driving power for each as a difference between the battery temperature T.sub.Batt and the predetermined low temperature T.sub.Low decreases.

[0088] In alternate embodiments, one or both of the side walls **30**, **32** can be omitted from the battery dock **20**. In alternate embodiments all of the walls **26**, **30**, **32** can be omitted from the battery dock **20**, and the cooling fan **62** and the dock opening **54** can be relocated in the support wall **28**.

[0089] In alternate embodiments, the battery charging system **10** can include a power button that is actuated by the user, and the controller **64** can be configured to detect the presence or absence of the battery **12** when the controller receives a signal from the power button that is indicative of the user's request for charging the battery **12**. The controller **64** can be configured to enter the battery heating and charging algorithm at step **S100** when the controller **64** receives an appropriate signal from the power button.

[0090] In alternate embodiments, the battery dock **20** can include a hollow compartment into which the battery **12** can be inserted. At least a portion of the compartment can be exposed to the ambient environment so that air can be exhausted from the battery **12** into the ambient environment during the heating mode and air can be drawn from the ambient environment and into the battery **12** during the cooling mode.

[0091] Although FIG. **4** shows the controller **64** in the interior space **60** of the housing **18**, alternate

embodiments can include the controller **64** located anywhere inside or outside of the housing **18** or inside the battery dock **20**. In alternate embodiments, the controller **64** can be enclosed in a compartment that is separated from the housing **18** and the battery dock **20**.

[0092] Although FIGS. **1-9** show the electronic component **76** in the battery compartment **84**, the electronic component **76** can be located anywhere in or on the battery **12** and can be located separate from the battery compartment **84**.

[0093] The housing opening **68** can be in any of the walls **34, 46, 48, 50** instead of, or in addition to, the first side wall **46**.

[0094] In alternate embodiments, the surfaces **36, 42** can abut each other.

Claims

1. A battery charging system, comprising: a heating element located in an airflow path; a heating fan located in the airflow path upstream of the heating element when the heating fan is operated during a heating mode; a cooling fan located in the airflow path downstream of the heating element when the heating fan is operated during the heating mode; and a controller configured to operate the heating mode when a battery is connected to the battery charging system and a sensed temperature of the battery is less than a predetermined low threshold, the controller is configured to operate the heating mode by disconnecting or maintaining disconnection of a charging power from the battery, supplying or maintaining a fan power to the heating fan to force air to flow along the airflow path, supplying or maintaining a heat power to the heating element to emit heat into the airflow path, and disconnecting or maintaining disconnection of a cooling power from the cooling fan, wherein in the heating mode, air flowing along the airflow path flows through the heating fan and enters the heating element, and the air that enters the heating element exits the heating element and passes along the cooling fan.
2. The battery charging system according to claim 1, further comprising: a battery dock configured to receive the battery when the battery is connected to the battery charging system, the battery dock is located such that the airflow path passes along the battery dock downstream of the heating element in the heating mode.
3. The battery charging system according to claim 2, wherein the controller is configured to operate a charging mode when the battery is connected to the battery charging system, and when the sensed temperature is greater than or equal to the predetermined low threshold and less than or equal to a predetermined high threshold, the predetermined high threshold is greater than the predetermined low threshold, the controller is configured to operate the charging mode by connecting or maintaining connection of the charging power to the battery, disconnecting or maintaining disconnection of the fan power from the heating fan, disconnecting or maintaining disconnection of the heat power from the heating element, and disconnecting or maintaining disconnection of the cooling power from the cooling fan.
4. The battery charging system according to claim 3, wherein the controller is configured to operate a cooling mode when battery is connected to the battery charging system and the sensed temperature is greater than the predetermined high threshold, the controller is configured to operate the cooling mode by connecting or maintaining connection of the charging power to the battery, disconnecting or maintaining disconnection of the fan power from the heating fan, disconnecting or maintaining disconnection of heat power from the heating element, and supplying or maintaining the cooling power to the cooling fan to force air to flow along the airflow path, and in the cooling mode, air flowing along the airflow path flows through the battery and subsequently flows past the heating element and the heating fan when the battery is connected to the battery charging system.
5. The battery charging system according to claim 4, further comprising: a housing that includes an interior space and a first opening, the first opening and at least a portion of the interior space are in the airflow path, the first opening is in fluid communication with an ambient environment outside

of the housing, wherein the battery dock includes an internal space and a second opening in the airflow path, the internal space is in fluid communication with the interior space, and the second opening is in fluid communication with the internal space, and the heating element, the heating fan, and the cooling fan are located along the airflow path between the first opening and the second opening, the heating element and the heating fan are located in the interior space of the housing, the cooling fan is located in the internal space of the battery dock, in the heating mode, the heating fan draws air from outside the housing into the interior space of the housing via the first opening, the air drawn into the interior space by the heating fan passes through the heating element and exits the internal space of the battery dock via the second opening, and in the cooling mode, the cooling fan draws air from the battery when the battery is received by the battery dock, the air drawn by the cooling fan enters the second opening, passes through the internal space and the interior space, and exits the housing via the first opening.

6. The battery charging system according to claim 1, further comprising: a battery dock configured to receive a battery; a first compartment in the battery dock, the cooling fan is contained in the first compartment; a second compartment adjacent to the first compartment, each of the controller and the heater are contained in the second compartment; and a first opening located in the airflow path and in fluid communication with each of the battery and the first compartment, a second opening located in the airflow path and in fluid communication with each of the first compartment and the second compartment, and a third opening located in the airflow path and in fluid communication with each of the second compartment and an ambient environment outside of the second compartment.

7. The battery charging system according to claim 1, wherein the controller is configured to, be connected to a power supply that is external to both the battery charging system and the battery that is to be charged by the battery charging system; selectively convert power received from the power supply to a charging power when the controller sets one of the charging mode and the cooling mode, and transmit the charging power to the electrical connector.

8. The battery charging system according to claim 7, wherein the controller is configured to electrically connect the heating element to the power supply when the controller sets the heating mode.

9. The battery charging system according to claim 8, wherein, when the controller sets the heating mode, the controller is configured to, calculate a heater power output to be emitted by the heating element based on the sensed temperature, and cause the heater element to emit heat at the heater power output.

10. The battery charging system according to claim 9, further comprising: a display screen that is electrically connected to the controller, wherein the controller is configured to, calculate an estimated wait time that corresponds to a difference between a start time of setting the heating mode and an estimated time of setting the charging mode, and cause the display screen to display the estimated wait time and the heater power output.

11. The battery charging system according to claim 1, wherein the controller is configured to electrically connect the battery to the heater when the controller sets the heating mode.

12. A battery power system for an apparatus that includes an electric motor and a tool driven by the electric motor, the battery charging system comprising: a battery configured to be removably mounted on the apparatus and electrically connected to the apparatus to supply electrical power to the motor when the battery is mounted on the apparatus; and a battery charging system connected to a power supply and including: a battery dock configured to removably receive the battery and electrically connect the battery to the battery charging system when the battery is received in the battery dock; a heater located in an airflow path that passes along the battery dock and is configured to heat the battery when the battery is received in the battery dock; a cooling fan located in an airflow path between the battery dock and the heater; and a controller configured to operate a heating mode when the battery is received in the battery dock and a sensed temperature is less than

a predetermined low threshold, the controller configured to operate the heating mode by disconnecting or maintaining disconnection of a charging power from the battery, connecting or maintaining connection of a heat power to the heater, and disconnecting or maintaining disconnection of a cooling power from the cooling fan.

13. The battery charging system according to claim 12, wherein the heater includes: a heating fan that is located in the airflow path at a position that is spaced away from the cooling fan; and a heating element that is located in the airflow path at a position that is between the heating fan and the cooling fan.

14. The battery charging system according to claim 12, wherein the controller is configured to operate a charging mode when the battery is received by the battery dock and when the sensed temperature is greater than or equal to the predetermined low threshold and less than or equal to a predetermined high threshold, the predetermined high threshold is greater than the predetermined low threshold, the controller is configured to operate the charging mode by connecting or maintaining connection of the charging power to the battery, disconnecting or maintaining disconnection of the heat power from the heater, and disconnecting or maintaining disconnection of the cooling power from the cooling fan.

15. The battery charging system according to claim 12, wherein the quick charger includes a housing containing the heater, and the housing includes a first opening in fluid communication with an ambient environment outside of the housing, the battery dock is adjacent to the housing and includes an external space and a second opening that is in fluid communication with the external space, the airflow path extends from the first opening, through the housing, through the second opening, and into the external space, and the battery is located in the external space when the battery is received by the battery dock, and the battery includes a ventilation opening that is in fluid communication with one of the external space and the second opening, the cooling fan is mounted in the battery dock at a location in the airflow path that is between the second opening and the heater, and the heater is mounted in the housing at a location in the airflow path that is between the cooling fan and the first opening.

16. The battery charging system according to claim 12, wherein the battery includes a plurality of battery cells and a plurality of thermistors, and a respective one of the thermistors abuts a respective one of the battery cells and is configured to output a respective temperature data signal to the controller, the temperature data includes the temperature signal output by each of the thermistors, and the controller is configured to set the sensed temperature based on the temperature data.

17. The battery charging system according to claim 16, wherein the plurality of battery cells are arranged in one of a row and a column and include, an end battery cell located at an end of the one of the row and the column, and a middle battery cell spaced away from the end of the one of the row and the column, and the plurality of thermistors includes a first thermistor abutting the end battery cell and a second thermistor abutting the middle battery cell.

18. A method of using a quick charger to charge a battery mounted in the quick charger, the quick charger having a heater, a cooling fan, and an opening adjacent to the battery when the battery is mounted in the quick charger, the method comprising: locating the heater, the cooling fan and the opening in an airflow path that extends from the heater and through the opening such that the cooling fan is located between the heater and the opening; and when a sensed temperature of the battery is less than a predetermined low threshold, disconnecting or maintaining disconnection of the battery from a power supply; connecting or maintaining connection of the heater to one of the power supply and the battery, and disconnecting or maintaining disconnection of the cooling fan from the power supply.

19. The method according to claim 18, further comprising: when the sensed temperature is greater than or equal to the predetermined low threshold and less than or equal to a predetermined high threshold that is greater than the predetermined low threshold, connecting or maintaining

connection of the battery to the power supply, disconnecting or maintaining disconnection of the cooling fan from the power supply, and disconnecting or maintaining disconnection of the heater from the one of the power supply and the battery; and when the sensed temperature is greater than the predetermined high threshold, connecting or maintaining connection of the battery to the power supply, connecting or maintaining connection of the cooling fan to the power supply, and disconnecting or maintaining disconnection of the heater from the one of the power supply and the battery.

20. The method according to claim 19, further comprising: setting the sensed temperature based on a plurality of temperature signals; varying a heater power output by the heater based on the sensed temperature; calculating an estimated wait time until a start of charging of the battery; and displaying the wait time and the heater power on a display.
